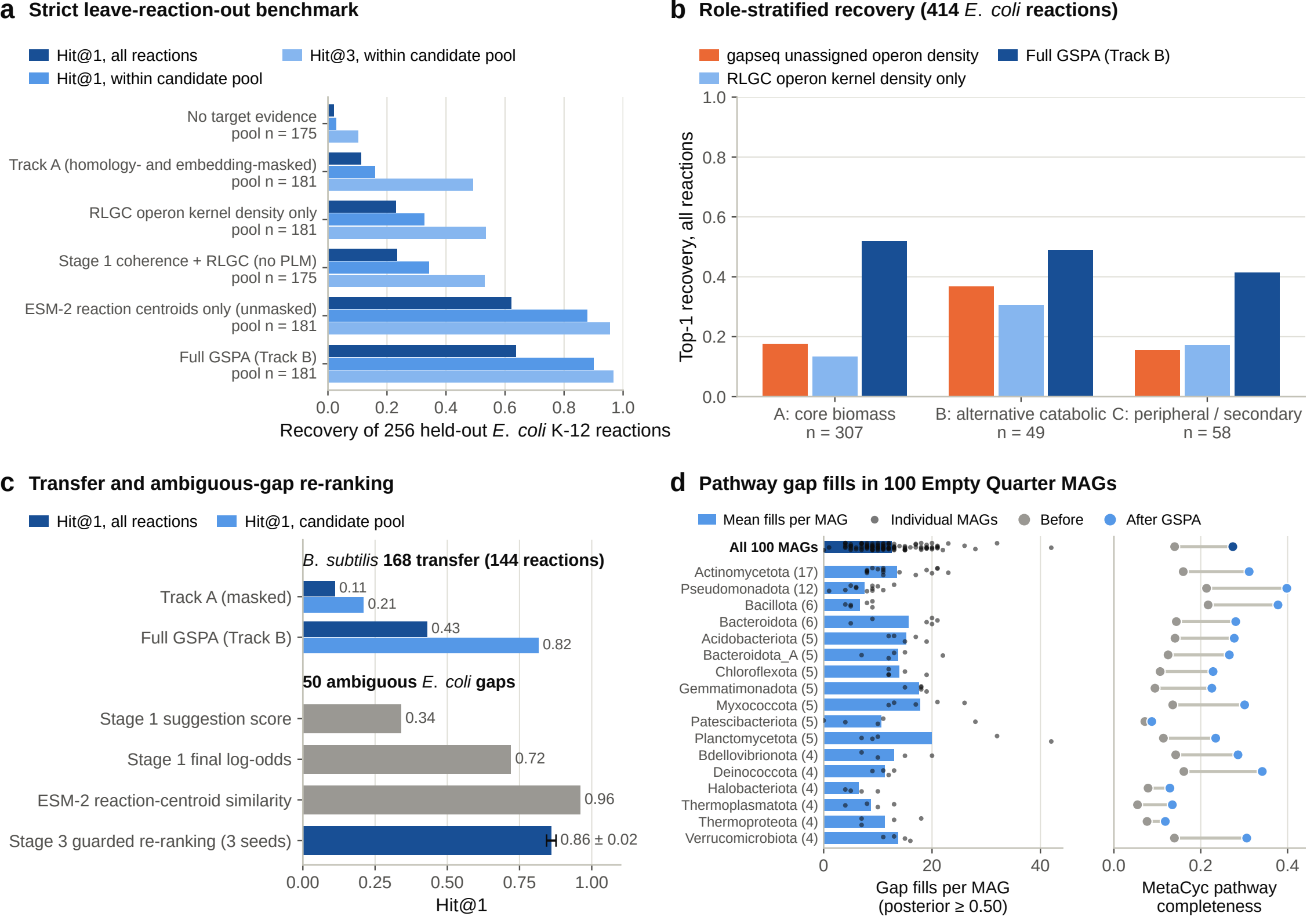
a Strict leave-reaction-out benchmark
Hit@1, all reactions
Hit@3, within candidate pool
Hit@1, within candidate pool
No target evidence pool n = 175
Track A (homology- and embedding-masked) pool n = 181
RLGC operon kernel density only pool n = 181
Stage 1 coherence + RLGC (no PLM) pool n = 175
ESM-2 reaction centroids only (unmasked) pool n = 181
Full GSPA (Track B) pool n = 181
0.0 0.2 0.4 0.6 0.8 1.0
Recovery of 256 held-out E. coli K-12 reactions
b Role-stratified recovery (414 E. coli reactions)
gapseq unassigned operon density
Full GSPA (Track B)
RLGC operon kernel density only
Top-1 recovery, all reactions
1.0 0.8 0.6 0.4 0.2 0.0
A: core biomass n = 307
B: alternative catabolic n = 49
C: peripheral / secondary n = 58
c Transfer and ambiguous-gap re-ranking
Hit@1, all reactions
Hit@1, candidate pool
B. subtilis 168 transfer (144 reactions)
Track A (masked)
0.11
0.21
Full GSPA (Track B)
0.43
0.82
50 ambiguous E. coli gaps
Stage 1 suggestion score
0.34
Stage 1 final log-odds
0.72
ESM-2 reaction-centroid similarity
0.96
Stage 3 guarded re-ranking (3 seeds)
0.86 ± 0.02
0.00 0.25 0.50 0.75 1.00
Hit@1
d Pathway gap fills in 100 Empty Quarter MAGs
Mean fills per MAG
Individual MAGs
Before
After GSPA
All 100 MAGs
Actinomycetota (17)
Pseudomonadota (12)
Bacillota (6)
Bacteroidota (6)
Acidobacteriota (5)
Bacteroidota_A (5)
Chloroflexota (5)
Gemmatimonadota (5)
Myxococcota (5)
Patescibacteriota (5)
Planctomycetota (5)
Bdellovibrionota (4)
Deinococcota (4)
Halobacteriota (4)
Thermoplasmatota (4)
Thermoproteota (4)
Verrucomicrobiota (4)
0 20 40
Gap fills per MAG (posterior ≥ 0.50)
0.0 0.2 0.4
MetaCyc pathway completeness